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微分几何

0. The local theory of curves

Def: A curve $\vec{r}(t): (a,b) \longrightarrow \mathbb{R}^n$ where $\vec{r}(t)=(x_1(t), \cdots, x_n(t))$ is regular if:

1° $x_i(t) \in C^\infty(a,b)$

2° $\forall t \in (a,b)$, $|\vec{r}'(t)| \neq 0$

Specifically, we often take $n=2,3$.

Notation: $\vec{r}' = \frac{d}{dt}\vec{r}(t)$ $\dot{\vec{r}}(s) = \frac{d}{ds}\vec{r}(s)$ for a regular curve $\vec{r}$ where s is called the arc length of $\vec{r}$. defined by the following formula.

$s = \int_a^t |\vec{r}'(u)|\,du$

Hence, we have $\frac{ds}{dt} = |\vec{r}'(t)|$

and $\dot{\vec{r}}(s) = \frac{d\vec{r}}{ds} = \frac{d\vec{r}}{dt}\cdot\frac{dt}{ds} = \vec{r}'\cdot\frac{1}{|\vec{r}'|} \Rightarrow \underline{\forall s,\ |\dot{\vec{r}}(s)|=1}$

Notation: Let $T(s) = \dot{\vec{r}}(s)$ be the tangent vector.

$N(s)$ be the normal vector with $|N|=1$

We assume that T is rotated counterclockwise by $\frac{\pi}{2}$ to obtain N.

• Plane curves $(n=2)$

Let $\vec{\gamma} = \vec{\gamma}(s)$. $T(s) = \dot{\vec{\gamma}}(s)$ and $|T(s)|=1$

then $|T(s)|^2 = 1 \Longrightarrow \frac{d}{ds}|T(s)|^2 = 0$ i.e. $\frac{d}{ds}\langle T(s), T(s)\rangle = 0$

$\Rightarrow 2\langle T(s), \dot{T}(s)\rangle = 0 \Rightarrow \langle \dot{T}(s), T(s)\rangle = 0$ i.e. $\underline{\dot{T}(s) = K_r(s)N(s)}$

where K_r is some <u>function</u> called the <u>relative curvature</u> of $\vec{\gamma}$.

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Lem. $\frac{d}{ds}\begin{pmatrix}T\\N\end{pmatrix} = \begin{pmatrix}0 & K_r\\ -K_r & 0\end{pmatrix}\begin{pmatrix}T\\N\end{pmatrix}$

Proof: $\dot{T}(s) = K_r \cdot N(s)$, since $|N(s)| = 1 \Rightarrow 0 = \langle \dot{N}, N\rangle$

$\frac{d}{ds}\langle T, N\rangle = 0 \Rightarrow \langle \dot{T}, N\rangle + \langle T, \dot{N}\rangle = 0$

$\Rightarrow K_r \langle N, N\rangle + \langle T, \dot{N}\rangle = 0 \Rightarrow \langle T, \dot{N}\rangle = -K_r$

i.e. $\dot{N}(s) = -K_r T(s)$ □

e.g. Given $\vec{r}(t) = (x_0 + R\cos t,\ y_0 + R\sin t)$ for $t \in (0, \pi)$, $R \in \mathbb{R}^+$, $(x_0, y_0) \in \mathbb{R}^2$

find $K_r(s)$.

Hint: $\vec{r}'(t) = R(-\sin t, \cos t) \Rightarrow |\vec{r}'(t)| = R = \frac{ds}{dt} \Rightarrow s = Rt$

Hence $\vec{r}(s) = (x_0 + R\cos\frac{s}{R},\ y_0 + R\sin\frac{s}{R})$

$T(s) = \dot{\vec{r}}(s) = (-\sin\frac{s}{R}, \cos\frac{s}{R})$

$\dot{T}(s) = \frac{1}{R}(-\cos\frac{s}{R}, -\sin\frac{s}{R})$ $\quad N(s) = (-\cos\frac{s}{R}, -\sin\frac{s}{R})$

$\Rightarrow K_r(s) = \frac{1}{R}$

e.g. $K_r(t) = \frac{x'y'' - x''y'}{(x'^2 + y'^2)^{3/2}}$ for a parameterized curve $\vec{r}(t) = (x(t), y(t))$

- Space curves $(n=3)$

Let $\vec{r}(s) = (x_1(s), x_2(s), x_3(s))$ be a regular curve, s.o.I

Def: The curvature of $\vec{r}(s)$ is $K(s) = |\ddot{\vec{r}}(s)|$

$\rho(s) = \frac{1}{K(s)}$ is called the radius of curvature

$|\dot{\vec{r}}(s)| = 1$ i.e. $\langle \dot{\vec{r}}(s), \dot{\vec{r}}(s)\rangle = 1 \Rightarrow \langle \ddot{\vec{r}}(s), \dot{\vec{r}}(s)\rangle = 0 \quad \forall s.$

W.L.O.G. $K(s) \neq 0$ $\quad N(s) = \frac{\ddot{\vec{r}}(s)}{|\ddot{\vec{r}}(s)|} = \frac{\ddot{\vec{r}}(s)}{K(s)}$

$\rightsquigarrow \dot{T}(s) = \ddot{\vec{r}}(s) = K(s)N(s)$, $\rightsquigarrow \dot{T} = KN$

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Def: $B(s) = T(s) \times N(s)$ is called binormal vector of $\vec{\gamma}(s)$

Hence we have an orthogonal frame along $\vec{\gamma}(s)$

called Frenet frame: $\{\vec{\gamma}(s); T, N, B\}$.

Def: The torsion of $\vec{\gamma}(s)$ is $\tau(s) = \langle \dot{N}, B \rangle$

Rmk: $\langle \dot{N}, B \rangle = \langle \dot{N}, T \times N \rangle = -\langle \dot{N}, N \times T \rangle = -\langle N, T \times \dot{N} \rangle$

since $\langle N, \dot{B} \rangle = \langle \dot{T} \times N, N \rangle + \langle T \times \dot{N}, N \rangle$

$= \langle \kappa N \times N, N \rangle + (-\langle \dot{N}, B \rangle)$

$= -\langle \dot{N}, B \rangle$

Hence $\underline{\tau(s) = \langle \dot{N}, B \rangle = -\langle N, \dot{B} \rangle}$

Thm. (Secret - Frenet formulas)

Let $\vec{\gamma}(s)$, $s \in I$ be a regular space curve.

Then.

$$\frac{d}{ds}\begin{pmatrix} T \\ N \\ B \end{pmatrix} = \begin{pmatrix} 0 & \kappa & 0 \\ -\kappa & 0 & \tau \\ 0 & -\tau & 0 \end{pmatrix}\begin{pmatrix} T \\ N \\ B \end{pmatrix}$$

Proof: We have $\langle \dot{T}, T \rangle = \langle \dot{N}, N \rangle = \langle \dot{B}, B \rangle = 0$ and $\dot{T} = \kappa N$

since $\langle T, N \rangle = 0 \Rightarrow \langle \dot{N}, T \rangle + \langle N, \dot{T} \rangle = 0$

$\Rightarrow \langle \dot{N}, T \rangle = -\kappa$

$0 = \langle B, T \rangle \Rightarrow 0 = \langle \dot{B}, T \rangle + \langle B, \dot{T} \rangle = \langle \dot{B}, T \rangle + \langle B, \kappa N \rangle$

Hence $\langle \dot{B}, T \rangle = 0$ □

Rmk: Note that $\dot{T}(s) = \ddot{\vec{\gamma}}(s) = \kappa(s) N(s)$, $\dot{N}(s) = -\kappa(s) T(s) + \tau(s) B(s)$

$\Rightarrow \dddot{\vec{\gamma}}(s) = \dot{\kappa} N + \kappa \dot{N} = \dot{\kappa} N - \kappa^2 T + \kappa \tau B$

It is worth noting that $\dot{N} = \frac{1}{\kappa}(\dddot{\vec{\gamma}} - \dot{\kappa} N)$

Hence $\tau = -\langle \dot{B}, N \rangle = \langle B, \dot{N} \rangle = \frac{1}{\kappa}\langle B, \dddot{\vec{\gamma}} \rangle - \langle B, N \rangle = \frac{1}{\kappa}\langle B, \dddot{\vec{\gamma}} \rangle$

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That is: $\tau(s) = \frac{1}{\kappa^2(s)} \langle \dot{\vec{\gamma}} \times \ddot{\vec{\gamma}}, \dddot{\vec{\gamma}} \rangle$

eg: The circular helix

$\vec{\gamma}(t) = (R\cos t, R\sin t, bt)$ $R \in \mathbb{R}_+$, $b \in \mathbb{R}$.

Prop A space curve is a plane curve iff $\tau(s) \equiv 0$ $\forall s \in I$

Proof: 1° "$\Rightarrow$" If $\vec{\gamma}(s)$, $s \in I$ is a plane curve

then $\exists\ \vec{n}_0 \in \mathbb{R}^3$ $|\vec{n}_0| = 1$, $x_0 \in \mathbb{R}^3$

s.t. $\langle \vec{n}_0, \vec{\gamma}(s) - x_0 \rangle = 0$

$\Rightarrow \langle \vec{n}_0, T(s) \rangle = 0 \Rightarrow \langle \vec{n}_0, \dot{\vec{\gamma}}(s) \rangle = 0$ i.e. $\langle \vec{n}_0, N(s) \rangle = 0$

$\rightsquigarrow \vec{n}_0 \perp T$, $\vec{n}_0 \perp N \Rightarrow \vec{n}_0 \parallel B \Rightarrow \tau = -\langle \dot{B}, N \rangle = 0$

2° "$\Leftarrow$" If $\tau(s) = 0$ $\forall s \in I$ then $0 = \tau(s) = -\langle \dot{B}, N \rangle$

$0 = \langle \dot{B}, B \rangle \Rightarrow \dot{B} \parallel T$ write $\dot{B} = \alpha T$

fact: $\langle \dot{B}, T \rangle = -\langle B, \dot{T} \rangle = -\langle B, \kappa N \rangle = 0$

$\Rightarrow \alpha = 0 \Rightarrow \dot{B}(s) = 0$ say $B(s) = \vec{n}_0$ for some $\vec{n}_0$, $|\vec{n}_0| = 1$

By definition $0 = \langle B(s), T(s) \rangle = \langle \vec{n}_0, \dot{\vec{\gamma}}(s) \rangle = \frac{d}{ds} \langle \vec{n}_0, \vec{\gamma}(s) - \vec{\gamma}(0) \rangle$

$\rightsquigarrow \vec{\gamma}(s)$, $s \in I$ is a plane curve. □

Def. A curve is called a helix

if the tangent lines of $\vec{\gamma}(s)$ make a const angle

with a fixed direction.

Lem. Assume $\tau(s) \neq 0$ $\forall s \in I$. Then $\vec{\gamma}$ is a helix

iff $\frac{\kappa(s)}{\tau(s)} = \text{const}$ $\forall s \in I$

Proof: 1° "$\Rightarrow$" If $\vec{\gamma}$ is a helix then $\exists\ \vec{a} \in \mathbb{R}^3$ $|\vec{a}| = 1$

$\langle \dot{\vec{\gamma}}(s), \vec{a} \rangle = \text{const} \Rightarrow \langle \ddot{\vec{\gamma}}, \vec{a} \rangle = 0$.

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$\Rightarrow \langle N(s), \vec{a} \rangle = 0 \quad \forall s.$

that is to say $a = \cos\theta\, T + \sin\theta\, B$ for some $\theta = \theta(s)$

then $0 = -\sin\theta \cdot \dot{\theta} \cdot T + \cos\theta\, \dot{T} + \cos\theta \cdot \dot{\theta} \cdot B + \sin\theta\, \dot{B}$

$= \dot{\theta}(-\sin\theta\, T + \cos\theta\, B) + \cos\theta\, \kappa N + \sin\theta\, (-\tau N)$

hence $\begin{cases} \dot{\theta} = 0 \Rightarrow \theta = \theta_0 \text{ for some const } \theta_0 \\ \kappa\cos\theta - \tau\sin\theta = 0 \Rightarrow \tan\theta = \dfrac{\kappa}{\tau} \end{cases}$

i.e. $\dfrac{\kappa}{\tau} = \tan\theta_0$. const

2° "$\Leftarrow$" By assumption $\dfrac{\kappa}{\tau} = \text{const} = \tan\theta$ (θ fixed)

Take $\vec{a}(s) = \cos\theta\, T + \sin\theta\, B$ cf. 1°

$\Rightarrow \vec{a}(s) = \vec{n}_0 \rightsquigarrow \vec{\gamma}(s)$ is a helix. □

Thm. (Fundamental Theorem for the local theory of curves)

Given smooth function $\kappa(s)$, $\tau(s)$ on (a, b) with $\kappa(s) > 0$

then there exists $\vec{\gamma}(s)$ has arc-length s

with curvature $\kappa(s)$ and torsion $\tau(s)$.

1. For $\vec{r}(t) \in E^3$ be a curve, $t \in I$

we have $\kappa(t) = \dfrac{|r' \times r''|}{|r'|^3}$ $\tau(t) = \dfrac{\langle r' \times r'', r''' \rangle}{|r' \times r''|^2}$

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